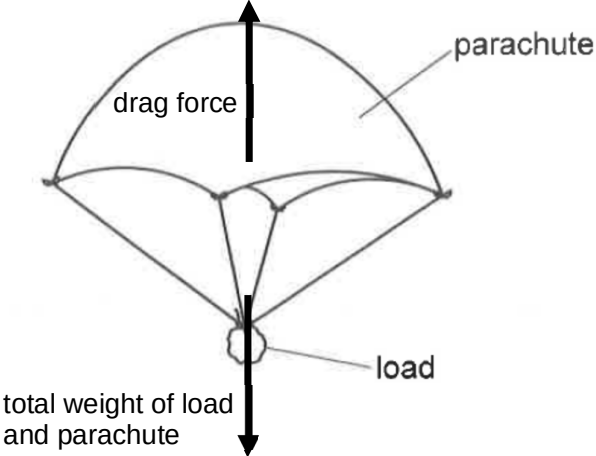
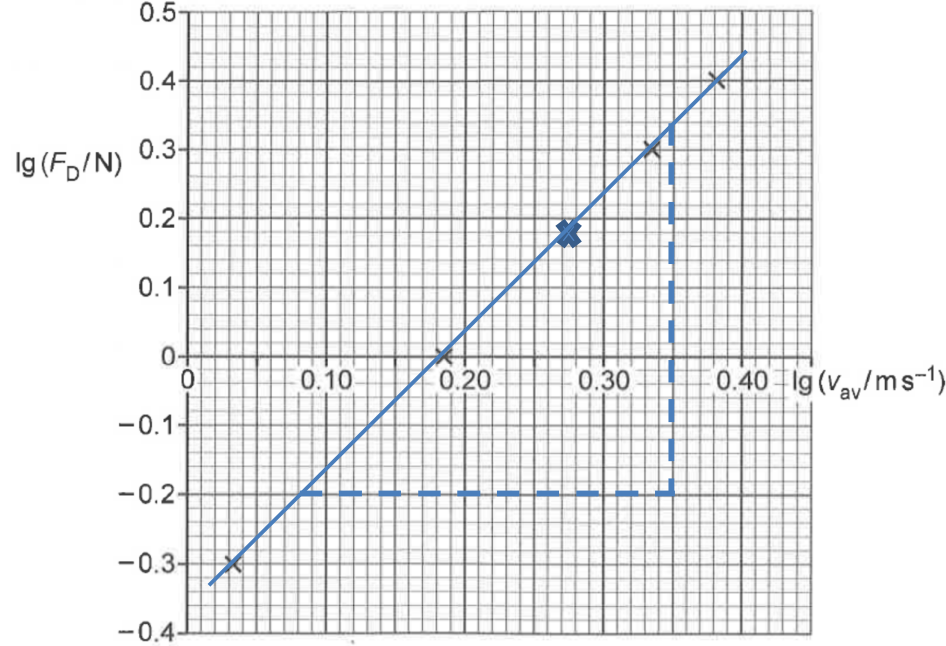


	as the initial velocity of object A.
2(a)(i)	SI base unit of drag force F_D is kg m s^{-2} SI base unit of speed v is m s^{-1} If the magnitude of the drag force depends only on the speed of the parachute, possible equations include $F_D = kv$ or $F_D = kv^2$ where the constant k is dimensionless. These equations will not be homogeneous.
2(a)(ii)	Surface area of parachute.

No.	Solution
2(b)(i)	
2(b)(ii)	Magnitude of drag force is equal to magnitude of total weight of load and parachute.
2(c)(i)	$v_1 = 2.47 \text{ m s}^{-1}$ $\lg F_D = 0.176$ $\lg v_{av} = 0.276$
2(c)(ii)	 Fig. 2.2

No.	Solution
2(c)(iii)	$F_D \propto (v_{av})^x$ $F_D = k (v_{av})^x$ $\lg F_D = x \lg v_{av} + \lg k$ where gradient = x and vertical intercept = $\lg k$ $\text{gradient} = \frac{0.34 - (-0.20)}{0.350 - 0.080} = \frac{0.54}{0.27} = 2.0$ $x = 2.0$
2(c)(iv)	vertical intercept = $\lg k$ $k = 10^{\text{vertical intercept}}$
2(c)(v)	$v_{av} = 1.11 \text{ m s}^{-1}$ $\lg v_{av} = 0.045$ (instead of 0.033) Plotted point shifts rightwards. Gradient of graph increases. Value of x increases.
3(a)	Period of satellite = 24 h